王保山，博士，副教授，硕导



wbs@ouc.edu.cn
bao-shan.github.io
0009-0007-8744-0648
中国海洋大学数学科学学院
山东青岛松岭路 238 号数学楼 402 室
Google Scholar, ResearchGate, Web of Science

研究方向

- 双曲守恒律方程高精度数值方法 (WENO 格式等) 研究
- 曲面双曲守恒律方程理论、方法与应用研究
- 多介质流模型高精度保物理约束数值方法研究

教育背景

2018.09 – 2022.06 **博士，中国海洋大学数学科学学院**计算数学专业
Wai Sun Don 教授
《保物理约束的高精度 WENO 格式》

2015.09 – 2018.06 **硕士，中国海洋大学数学科学学院**计算数学专业
高振教授，Wai Sun Don 教授
《基于径向基函数的间断检测算法及其在高精度杂交格式中的应用》

2011.08 – 2015.06 **学士，中国海洋大学数学科学学院**数学与应用数学专业

工作经历

2024.09 – 至今 青年英才工程**副教授**，**硕导**，中国海洋大学数学科学学院

2022.08 – 2024.09 **科研博士后**，中国海洋大学数学科学学院
高振教授
《多介质流体的高精度保物理约束 AWENO 格式》

学术访问

2026 香港浸会大学数学系，2026.01.04–01.15, 2026.09.01–09.19
Leevan Ling 教授 (数学系主任)　Wai Sun Don 教授

2025 韩国浦项科技大学数学系，访问教授，2025.12.25–2026.01.02
Jae-Hun Jung 教授 (理学院院长、数学系主任)　Jiaxi Gu 博士

2025 香港浸会大学数学系，2025.11.02–11.29
Leevan Ling 教授　Wai Sun Don 教授

2024 香港浸会大学数学系，2024.11.03–11.30
Leevan Ling 教授　Wai Sun Don 教授

2024 南方科技大学杰曼诺夫数学中心，2024.03.03–05.31
Alexander Kurganov 教授 (讲席教授，入选国家重要人才计划)

2024 香港科技大学数学系，2024.05.05–05.11
Zhichao Peng 教授

学术访问 (continued)

2023 香港浸会大学数学系，2023.10.16–11.06，2023.12.16–2024.01.09
Leevan Ling 教授　　Wai Sun Don 教授

2023 南方科技大学杰曼诺夫数学中心，2023.04.03–09
Alexander Kurganov 教授

2019 南方科技大学数学系，2019.03.27–04.01，09.15–19，12.17–20
Alexander Kurganov 教授

研究成果 (* = 通讯作者; ♯ = 同等贡献; IF = 影响因子; 累积被引 = 496; H-Index = 13)

1. **Bao-Shan Wang***. On wave speeds in the HLLC Riemann solver for the compressible two-medium flows with general equations of state. *Journal of Computational Physics*, 2027, 569(2), 115501. (IF = 3.9)
2. Khaled Bensayah, **Bao-Shan Wang***, Jia-Hao Liu, & Abdellah Hadjadj. WENO-ZKB: An improved WENO-Z+ scheme with new anti-dissipative term. *Journal of Computational Physics*, 2027, 569(1), 115434. (IF = 3.9)
3. Changming Guo♯, **Bao-Shan Wang**♯, Wai Sun Don♯, & Yuanyang Qiao*,♯. High-order physical-constraints-preserving velocity-consistent schemes for compressible multicomponent five-equation model with the Mie-Grüneisen equation of state. *Journal of Computational Physics*, 2026, 563, 115049. (IF = 3.9)
4. Ya-Ru Zhao, Zhen Gao, & **Bao-Shan Wang***. Fifth-order equilibrium-preserving path-conservative characteristic-wise AWENO scheme for one-fluid two-temperature Euler model. *Journal of Computational Physics*, 2026, 544, 114409. (IF = 3.9)
5. Xiao-Shuo Xiang, Peng Li, Zhen Gao, & **Bao-Shan Wang***. High-order well-balanced positivity-preserving affine-invariant finite volume CWENOZ scheme with LDCU flux for the Euler equations under gravitational fields, *Applied Numerical Mathematics*, 2026, 224, 181–206. (IF = 2.4)
6. **Bao-Shan Wang**, Alex Shiu Lun Chu, Leevan Ling, & Wai Sun Don*. Foliation structures and global flow dynamics of scalar hyperbolic conservation laws on manifolds: I. Geometry-compatible fluxes and numerical validation on sphere and torus. *Communications in Nonlinear Science and Numerical Simulation*, 2026, 152, 109415. (IF = 3.8)
7. Xiao-Shuo Xiang, Peng Li, & **Bao-Shan Wang***. Finite-volume WENO scheme on non-uniform mesh for 3D detonation simulation. *Applied Mathematics Letters*, 2026, 182, 110046. (IF = 2.7)
8. Kang-Bo Tian, **Bao-Shan Wang**, Xiao Wen, & Zhen Gao*. High order well-balanced and positivity-preserving AWENO scheme for rotating shallow water equations with Coriolis force. *East Asian Journal on Applied Mathematics*, 2026, 16(4), 714–741. (IF = 1.1)
9. Wai Sun Don♯, Jia-Le Li♯, Leevan Ling♯, **Bao-Shan Wang***,♯, & Yinghua Wang♯. Hybrid high-order shock-capturing scheme for one-dimensional hyperbolic conservation laws on manifolds (surface PDEs) in the time-continuous embedding framework. in: S. Chun, J.-H. Jung, E.J. Park, J. Shen (eds), *Spectral and High-Order Methods for Partial Differential Equations ICOSAHOM 2023. Lecture Notes in Computational Science and Engineering*, vol. 142, Springer, Cham, 2025, pp. 239–255.
10. **Bao-Shan Wang** & Wai Sun Don*. Fifth-order bound-, positivity-, and equilibrium-preserving affine-invariant AWENO scheme for two-medium γ-based model of stiffened gas. in: R.S. Myong and H.D. Kim (eds), *Proceedings of the 34th International Symposium on Shock Waves, Volume 1: Fundamentals*, Springer, Singapore, 2025, pp. 59-71.
11. Qingcheng Fu, Yaguang Gu*, Alexander Kurganov, & **Bao-Shan Wang**. Bound- and positivity-preserving path-conservative central-upwind AWENO scheme for the five-equation model of compressible two-component flows. *Journal of Scientific Computing*, 2025, 104(3), 94. (IF = 3.3)

12. Changming Guo, Yuanyang Qiao, Wai Sun Don, & **Bao-Shan Wang***. A family of bound-preserving velocity-consistent schemes for two-medium γ-based model with stiffened gas. *Journal of Computational Physics*, 2025, 538, 114149. (IF = 3.8)

13. Cai-Feng Wang♯, Wai Sun Don♯, Jia-Le Li♯, & **Bao-Shan Wang***,♯. Improved sixth-order WENO finite difference schemes for hyperbolic conservation laws. *Advances in Applied Mathematics and Mechanics*, 2025, 17(6), 1591–1624. (IF = 1.1)

14. Jia-Le Li, Wai Sun Don, Cai-Feng Wang, & **Bao-Shan Wang***. Spatial-temporal adaptive-order positivity-preserving WENO finite difference scheme with relaxed CFL condition for Euler equations with extreme conditions. *Advances in Applied Mathematics and Mechanics*, 2025, 17(3), 804–839. (IF = 1.5)

15. Zhen Gao, Zi-Yu Tang, **Bao-Shan Wang***, & Ya-Ru Zhao. Novel high-order alternative finite difference central WENO schemes for hyperbolic conservation laws. *Numerical Methods for Partial Differential Equations*, 2025, 41(2), e23170. (IF = 2.1. 凭此文获 2025 年第一季度 Wiley 威立中国高贡献作者奖)

16. Xiao-Shuo Xiang♯, **Bao-Shan Wang**♯, Zhen Gao♯, & Peng Li*,♯. Affine-invariant WENO operator on nonuniform Cartesian mesh with application to finite volume and discontinuous Galerkin methods. *Journal of Scientific Computing*, 2025, 102(2), 29. (IF = 2.8)

17. Baifen Ren, **Bao-Shan Wang**, Xiangxiong Zhang, & Zhen Gao*. Positivity and bound preserving well-balanced high order compact finite difference scheme for Ripa and pollutant transport model. *Computers and Mathematics with Applications*, 2024, 176, 545–563. (IF = 2.9)

18. Zhen Gao♯, Shuang Guo♯, **Bao-Shan Wang**♯, & Yaguang Gu*,♯. High order bound- and positivity-preserving finite difference affine-invariant AWENO scheme for the five-equation model of two-medium flows. *Communications in Computational Physics*, 2024, 36(3), 781–820. (IF = 2.6)

19. **Bao-Shan Wang** & Naveen Kumar Garg*. Third-order numerical scheme for Euler equations of gas dynamics using Jordan canonical based splitting flux. *Computers & Fluids*, 2024, 281, 106370. (IF = 2.5)

20. Zhen Gao, Ya-Ru Zhao, & **Bao-Shan Wang***. Sixth-order perturbed WENO interpolation-based AWENO and WCNS-E schemes for hyperbolic conservation laws. *Applied Mathematics Letters*, 2024, 158, 109230. (IF = 2.9)

21. Qingcheng Fu♯, Zhen Gao♯, Yaguang Gu♯, Peng Li*,♯, & **Bao-Shan Wang**♯. Improved well-balanced AWENO schemes with hydrostatic reconstruction for the Euler equations under gravitational fields. *Mathematics and Computers in Simulation*, 2024, 221, 260–280. (IF = 4.6)

22. Peng Li, Tingting Li, Wai Sun Don, & **Bao-Shan Wang***. Scale-invariant multi-resolution alternative WENO scheme for the Euler equations. *Journal of Scientific Computing*, 2023, 94(1), 15. (IF = 2.5)

23. Yinghua Wang, Wai Sun Don, & **Bao-Shan Wang***. Fifth order AWENO finite difference scheme with adaptive numerical diffusion for Euler equations. *Computers & Fluids*, 2023, 251, 105743. (IF = 3.077)

24. Kang-Bo Tian, Wai Sun Don, & **Bao-Shan Wang***. High order WENO finite difference scheme with adaptive dual order ideal weights for hyperbolic conservation laws. *Applied Numerical Mathematics*, 2023, 187, 50–70. (IF = 2.994. 2023 年 10 月 25 日在过去 90 天最高下载文章中排第 16 名)

25. **Bao-Shan Wang**, Wai Sun Don, & Peng Li*. Fifth-order well-balanced positivity-preserving finite difference AWENO scheme with hydrostatic reconstruction for hyperbolic chemotaxis models. *Applied Numerical Mathematics*, 2023, 186, 41–56. (IF = 2.994)

26. **Bao-Shan Wang**, Wai Sun Don, Alexander Kurganov*, & Yongle Liu. Fifth-order A-WENO schemes based on the adaptive diffusion central-upwind Rankine-Hugoniot fluxes. *Communications on Applied Mathematics and Computation*, 2023, 5, 295–314. (IF = 1.6)

27 Yinghua Wang♯, **Bao-Shan Wang**♯, Leevan Ling♯, & Wai Sun Don*,♯. A time-continuous embedding method for scalar hyperbolic conservation laws on manifolds. *Journal of Scientific Computing*, 2022, 93(3), 84. (IF = 2.5)

28 **Bao-Shan Wang**, Peng Li, & Zhen Gao*. High order well-balanced and positivity-preserving finite difference AWENO scheme with hydrostatic reconstruction for shallow water equations. *Applied Numerical Mathematics*, 2022, 181, 483–502. (IF = 2.994)

29 **Bao-Shan Wang** & Wai Sun Don*. Affine-invariant WENO weights and operator. *Applied Numerical Mathematics*, 2022, 181, 630–646. (IF = 2.994)

30 Wai Sun Don♯, Run Li♯, **Bao-Shan Wang**♯, & Yinghua Wang*,♯. A novel and robust scale-invariant WENO scheme for hyperbolic conservation laws. *Journal of Computational Physics*, 2022, 448, 110724. (IF = 4.645)

31 Peng Li♯, **Bao-Shan Wang**♯, & Wai Sun Don*,♯. Sensitivity parameter-independent characteristic-wise well-balanced finite volume WENO scheme for the Euler equations under gravitational fields. *Journal of Scientific Computing*, 2021, 88(2), 47. (IF = 2.843)

32 Zhen Gao*, Qi Liu, Jan S. Hesthaven, **Bao-Shan Wang**, Wai Sun Don, & Xiao Wen. Non-intrusive reduced order modeling of convection dominated flows using artificial neural networks with application to Rayleigh-Taylor instability. *Communications in Computational Physics*, 2021, 30(1), 97–123. (IF = 3.246)

33 **Bao-Shan Wang**, Wai Sun Don, Naveen Kumar Garg*, & Alexander Kurganov. Fifth-order A-WENO finite difference schemes based on a new adaptive diffusion central numerical flux. *SIAM Journal on Scientific Computing*, 2020, 42(6), A3932–A3956. (IF = 2.373)

34 Wai Sun Don, Dong-Mei Li, Zhen Gao, & **Bao-Shan Wang***. A characteristic-wise alternative WENO-Z finite difference scheme for solving the compressible multicomponent non-reactive flows in the overestimated quasi-conservative form. *Journal of Scientific Computing*, 2020, 82(2), 27. (IF = 2.592)

35 Zhen Gao, Li-Li Fang, **Bao-Shan Wang**, Yinghua Wang, & Wai Sun Don*. Seventh and ninth orders alternative WENO finite difference schemes for hyperbolic conservation laws. *Computers & Fluids*, 2020, 202, 104519. (IF = 3.013)

36 Peng Li, Xiqiang Zhao, Zhen Gao*, & **Bao-Shan Wang**. High order hybrid weighted compact nonlinear schemes for hyperbolic conservation laws. *Advances in Applied Mathematics and Mechanics*, 2020, 12(4), 972–991. (IF = 1.727)

37 Yinghua Wang♯, **Bao-Shan Wang**♯, & Wai Sun Don*,♯. Generalized sensitivity parameter free fifth order WENO finite difference scheme with Z-type weights. *Journal of Scientific Computing*, 2019, 81(3), 1329–1358. (IF = 2.228)

38 **Bao-Shan Wang**, Peng Li, Zhen Gao, & Wai Sun Don*. An improved fifth order alternative WENO-Z finite difference scheme for hyperbolic conservation laws. *Journal of Computational Physics*, 2018, 374, 469–477. (IF = 2.845)

39 **Bao-Shan Wang**, Wai Sun Don*, Zhen Gao, Yinghua Wang, & Xiao Wen. Hybrid Compact-WENO finite difference scheme with radial basis function based shock detection method for hyperbolic conservation laws. *SIAM Journal on Scientific Computing*, 2018, 40(6), A3699–A3714. (IF = 2.31)

40 Wai Sun Don♯, **Bao-Shan Wang**♯, & Zhen Gao*,♯. Fast iterative adaptive multi-quadric radial basis function method for edges detection of piecewise functions—I: uniform mesh. *Journal of Scientific Computing*, 2018, 75(2), 1016–1039. (IF = 2.370)

41 王立锋*, 叶文华, 陈竹, & 等. 激光聚变内爆流体不稳定性基础问题研究进展. 强激光与粒子束, 2021, 33(1), 012001.

42 杨洋, 曾维新, 高振, & **王保山***. 求解双曲守恒律的紧–WENO 杂交格式及 RBF–FD 间断检测方法. 数值计算与计算机应用, 2020, 41(3), 232–245.

43 徐捷, 高振, 曾维新, & **王保山***. 高阶驻点上精度保持的 WENO 有限差分格式. 数值计算与计算机应用, 2020, 41(1), 68–82.

44 **王保山**, 曾维新, 高振*, & 陈玉虎. Compact–WENO 杂交格式中旋涡识别的多分辨率分析方法. 中国海洋大学学报 (自然科学版), 2018, 48(增 II), 198–202.

已接受论文

1 Qingcheng Fu, Alexander Kurganov*, & **Bao-Shan Wang**. Local characteristic decomposition-based central-upwind flux splitting. *Journal of Scientific Computing*, 2026. (IF = 2.6)

2 Yuanyang Qiao♯, Yun-Xia Liu♯, **Bao-Shan Wang***,♯, & Wai Sun Don♯. Affine-invariant trigonometric WENO finite difference schemes for hyperbolic conservation laws. *Communications in Computational Physics*, 2025. (IF = 3.1)

3 Changming Guo♯, Yuanyang Qiao♯, **Bao-Shan Wang**♯, Wai Sun Don*,♯. Bound- and positivity-preserving first-order velocity-consistent HLL scheme for two-medium γ-based model of stiffened gas. in: D. E. Gildfind, R. Gollan and H. H. Kleine (eds), *Proceedings of the 35th International Symposium on Shock Waves: ISSW35, July 7-11 2025, Brisbane, Australia*.

已投稿论文

1 Shaoshuai Chu, Alexander Kurganov*, Mingye Na, **Bao-Shan Wang**, & Ruixiao Xin. Local characteristic decomposition of equilibrium variables for hyperbolic systems of balance laws. submitted to *Applied Numerical Mathematics*. under revision.

2 Ya-Ru Zhao, Zhen Gao, **Bao-Shan Wang***, & Wai Sun Don. High-order physical-constraints-preserving common-weights A-WENO scheme for non-conservative five-equation model with stiffened gas EOS. submitted to *Journal of Computational Physics*. under revision.

3 Alex Shiu Lun Chu♯, Wai Sun Don♯, Leevan Ling♯, & **Bao-Shan Wang***,♯. Foliation structures and global flow dynamics of scalar hyperbolic conservation laws on manifolds: II. An adaptively penalized closest point embedding with a WENO finite difference scheme (AP-cp-WENO). submitted to *SIAM Journal on Scientific Computing*.

4 Jia-Hao Liu, **Bao-Shan Wang**, & Ya-Ru Zhao*. Sixth-order AWENO schemes with adaptive linear weights for hyperbolic conservation laws. submitted to *Communications in Computational Physics*.

5 Changming Guo, Pengzhan Huang*, Yuanyang Qiao, **Bao-Shan Wang**. Accurate simulation of the granular avalanche flows by high-order positivity-preserving affine-invariant AWENO finite difference schemes. submitted to *Computers and Fluids*.

6 Yanping Qiu, Zhen Gao, Alexander Kurganov, **Bao-Shan Wang***, & Xiao Wen. Fifth-order well-balanced path-conservative A-WENO scheme for the Ripa model. submitted to *Journal of Computational Mathematics*.

7 Meifang Zhang♯, **Bao-Shan Wang**♯, Wai Sun Don♯, & Yuanyang Qiao*,♯. Spatially adaptive-order bound-preserving velocity-consistent AWENO schemes for compressible two-medium flows. submitted to *Journal of Scientific Computing*.

8 Hai-Liang Nie, Yuanyang Qiao, **Bao-Shan Wang***, & Wai Sun Don. RBF-AWENO scheme with affine-invariant weights: Achieving sixth-order accuracy and multiscale robustness. submitted to *Communications in Computational Physics*. under revision.

9. Ya-Ru Zhao, Zhen Gao, & **Bao-Shan Wang***. Physical-constraints-preserving path-conservative Lax-Friedrichs schemes for non-conservative two-medium six-equation model. submitted to *Journal of Scientific Computing*.
10. Naveen Kumar Garg*, Jhantu Pal, & **Bao-Shan Wang**. A Low-Dissipation LLF-Type Adaptive Central Scheme for the Compressible Euler Equations. submitted to *Mathematics and Computers in Simulation*.
11. Jiaxi Gu, **Bao-Shan Wang***, Wai Sun Don, & Jae-Hun Jung. A family of even-order central-upwind WENO schemes with averaged downwind and novel global smoothness indicators. submitted to *Numerical Methods for Partial Differential Equations*.
12. Xiao-Shuo Xiang, **Bao-Shan Wang***. Adaptive alternative weighted essentially non-oscillatory scheme for solving Degasperis-Procesi equation. submitted to *Advances in Applied Mathematics and Mechanics*.
13. Jing Li, Peng Li* & **Bao-Shan Wang**. Physical-constraints-preserving conservative-variable AWENO scheme for two-medium flows. submitted to *Advances in Applied Mathematics and Mechanics*.
14. 邱燕萍, 高振, 王保山*. 热旋转浅水波模型的五阶保平衡路径守恒特征 AWENO 格式. 已投稿至数值计算与计算机应用.

待整理论文

1. Shan Zhao, **Bao-Shan Wang**, & Peng Li*. Bound-Preserving Central-Upwind Scheme for One- and Multi-Dimensional Detonation Simulations
2. Kang-Bo Tian, **Bao-Shan Wang**, & Zhen Gao*. High-order well-balanced AWENO scheme for thermal rotating shallow water equations with Coriolis force.
3. Siyuan Lang, Zhiyue Zhang, **Bao-Shan Wang**, & Wai Sun Don. A physics-informed domain decomposition method with explicit shock tracking for 1D scalar conservation laws.

待出版著作

1. Wai Sun Don & **Bao-Shan Wang**. Lecture on the High-Order WENO Scheme for Hyperbolic Conservation Laws (HKBU edition). DOI:10.13140/RG.2.2.24752.37123/2.
2. Wai Sun Don & **Bao-Shan Wang**. Comprehensive Lecture Notes on Hyperbolic Conservation Laws on Manifolds. DOI:
3. **王保山**. 我的 MATLAB 报告. 整理中.

研究项目

2024.01 – 2026.12 国家自然科学基金青年项目/12301530/曲面双曲守恒律方程的时间连续最近点方法/30 万/**主持**

2025.01 – 2027.12 中国海洋大学青年英才工程科研启动费/862501013106/极端条件下多介质流模型的高精度保物理约束数值方法研究/30 万/**主持**

2026.01 – 2028.12 2025 年度山东省高等学校青年创新团队计划/多介质流模型数值模拟与应用创新团队/2025KJG007/多介质流模型保物理约束数值方法研究/**主持**

2023.01 – 2025.12 山东省自然科学基金面上项目/ZR2022MA012/可压缩多组分流模型基于物理结构的高精度自适应杂交求解器/10 万/**负责**/资助期满

2023.07 – 2024.07 中国博士后科学基金面上项目/2023M733348/Mie-Grüneisen 状态方程下两介质六方程模型的保界限制器研究/8 万/**主持**/结题

研究项目 (continued)

2024.01 – 2027.12 国家自然科学基金面上项目/12371435/基于深度神经网络的新型变分同化方法研究/43.5 万/参与/第五位

2024.01 – 2026.12 山东省自然科学基金面上项目/ZR2023MA077/反常扩散方程的局部误差估计与误差界爆破现象研究/10 万/参与/第二位

2024.01 – 2026.12 山东省自然科学基金面上项目/ZR2023MA043/数据驱动的南海内波高效数值模拟方法研究/10 万/参与/第五位

2024.01 – 2026.12 山东省高等学校青年创新团队发展计划/2023KJI020/海洋内波的数学分析与求解理论研究/0 万/参与/第六位

2022.10 – 2025.09 崂山实验室创新项目/LSKJ202202304/北极海冰预测系统的国产计算平台移植与优化/300 万/参与/第十位/结题

2019.01 – 2022.12 国家自然科学基金面上项目/11871443/不确定性下的三维气体爆炸波的数值方法研究/50 万/参与/第八位/结题

2017.01 – 2019.12 国家重大科技专项项目/***/*** 基础问题研究/90 万/参与/第八位/结题

教学经历

2026.03 – 2025.07 本科生课程《复变函数》(代码 008401101031, 100 + 100 人)

2026.03 – 2025.07 研究生课程《谱方法》(代码 100K0171001，3 人 + 2 人旁听)

2025.02 – 2025.07 本科生课程《复变函数》(代码 008401101031, 149 + 76 人)

2025.02 – 2025.07 研究生课程《谱方法》(代码 100K0171001，6 人 + 2 人旁听)

2021.09 – 2022.01 迷你课程：《科学计算与编程》、《非线性波动方程》

2020.03 – 2020.07 研究生课程《谱方法》

学术活动

2026.10.17 Spatially Adaptive-Order Bound-Preserving Velocity-Consistent AWENO Schemes for Compressible Two-Medium Flows, High-Order and Adaptive Schemes for Nonlinear Hyperbolic PDEs, SUSTech, Shenzhen

2026.08.27 Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds, EASIAM 2026, Jeju, South Korea

2026.08.18 High-order physical-constraints-preserving velocity-consistent schemes for compressible multicomponent five-equation model with the Mie-Grüneisen equation of state, 昆明理工大学，昆明，云南.

2026.01.30 浅水波类模型的高精度保平衡 AWENO 格式, 崂山国家实验室, 青岛

2026.01.17 Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds, 偏微分方程高精度数值算法及应用研讨会，广西大学，南宁

2026.01.07 —, Conference on Modern Kernel Methods and Applications, On the Occasion of Robert Schaback's 80th Birthday, HKBU

2025.10.26 A family of bound-preserving velocity-consistent schemes for two-medium γ-based model with stiffened gas, CSIAM 2025 年会，长沙，湖南

2025.08.09 —，第二十一届全国流体力学数值方法研讨会，伊宁，伊犁

2025.07.27 Bound- and positivity-preserving path-conservative central-ipwind AWENO scheme for the five-equation model of compressible two-component flows，新疆大学，乌鲁木齐

2024.12.31 —，河南工业大学，郑州

学术活动 (continued)

2024.11.11 A Time-Continuous Embedding Method for Hyperbolic Conservation Laws on Manifolds，香港浸会大学数学系，香港

2024.06.24 Fifth-order bound-, positivity-, and equilibrium-preserving affine-invariant AWENO scheme for two-medium γ-based model of stiffened gas，新疆大学，乌鲁木齐

2024.04.07 Time-Continuous Embedding Method for Scalar Hyperbolic Conservation Laws on Manifolds，石家庄铁道大学，石家庄

2024.03.19 Affine-invariant WENO operator and applications in solving hyperbolic conservation laws, Workshop on Development of High-Order Methods for Hyperbolic PDEs，南方科技大学杰曼诺夫数学中心，深圳

2023.11.11 Time-Continuous Embedding Method for Scalar Hyperbolic Conservation Laws on Manifolds，哈尔滨工业大学 (深圳)，深圳

2023.10.24 Affine-invariant WENO weights and applications in solving hyperbolic conservation laws，香港浸会大学，香港

2023.08.16 —，ICOSAHOM 2023，首尔

2023.04.04 —，杰曼诺夫数学中心，深圳

2023.04.02 —，第二十届全国流体力学数值方法研讨会，江苏南京

2022.07.12 仿射不变性的高精度低耗散 WENO 格式，新疆大学（线上）

2021.07.03 Alternative WENO Scheme with Adaptive Diffusion Numerical Flux for Hyperbolic Conservation Laws，山东省计算数学学术研讨会，山东烟台

2020.11.15 —，中国工业与应用数学学会学生论坛（线上）

2019.11.05 Non-intrusive reduced order modeling of convection dominated flows using artificial neural networks with application to Rayleigh-Taylor instability. International Joint Conference on AI & Data Science: Mathematics & Applications，韩国亚洲大学

2019.06.24 Hybrid compact-WENO finite difference scheme with radial basis function based shock detection method for hyperbolic conservation laws. The 11th International Conference on Computational Physics，浙江大学

2019.04.20 —，石家庄铁道大学

2018.12.07 烧蚀瑞利泰勒不稳定性数值模拟，北京应用物理与计算数学研究所

2017.08.09 Hybrid compact-WENO finite difference scheme with radial basis function based shock detection method for hyperbolic conservation laws，第十八届全国流体力学数值方法研讨会，怀化学院

专业服务

期刊审稿 (部分列表)

30 份 Journal of Computational Physics

26 份 Applied Numerical Mathematics

12 份 Advances in Applied Mathematics and Mechanics

9 份 Communications in Nonlinear Science and Numerical Simulation

6 份 Computers & Fluids

6 份 Journal of Computational and Applied Mathematics

4 份 Computers and Mathematics with Applications

2 份 Advances in Water Resources

专业服务 (continued)

1 份 Communications in Computational Physics

21 份 其他期刊

专题研讨会组织者

2026 EASIAM 2026, High-Order Numerical Methods for Hyperbolic Conservation Laws, 8 月 23–27 日, 韩国济州岛

2023 ICOSAHOM 2023, Recent developments in physical-property-preserving WENO schemes and DG methods for hyperbolic single- and multi-medium models, 8 月 14–18 日, 延世大学, 韩国首尔

学术兼职

2025– 骨干成员，山东省随机系统控制理论与科学计算重点实验室

2024– 骨干成员，海洋数学技术联合实验室

2022– 专业会员，中国工业与应用数学学会

获奖情况

2023 山东省优秀博士学位论文

2021 博士研究生国家奖学金、中国海洋大学第八届研究生年度人物称号 (共 10 项)

2019 山东省优秀硕士学位论文

2018 山东省优秀硕士毕业研究生

2017 硕士研究生国家奖学金

2016 研究生数学建模竞赛国家一等奖和赛事优秀论文

2015 研究生数学建模竞赛国家二等奖

2014 大学生数学建模竞赛国家二等奖

其他技能

创建微信公众号 “台劳公式”：分享科研经验、编程技术和快乐数学 (3200+ 关注)

熟练掌握软件：MATLAB 和 Fortran 编程，LaTeX 文档排版

熟练使用 Linux 操作系统，维护 Linux 服务器

指导学生情况

博士研究生 (协助指导)

2025 – 2029 赵亚茹 [在读]; 郭长明 (新疆大学) [在读]

2022 – 2027 项晓硕 [在读]

2021 – 2023 王海云, 可压缩多介质流数值模拟的保界保正间断 Galerkin 方法研究, 湘潭大学

2020 – 2026 田康博, 三类浅水波方程五阶保平衡保正 WENO 有限差分格式的研究, 青岛黄海学院

2020 – 2025 任佰芬, 浅水波方程组的保界保平衡高精度有限差分格式研究, 山东工商学院

指导学生情况 (continued)

硕士研究生

2026 – 2029 丁佳旭, 赵杉 [在读]

2025 – 2028 刘家豪 [在读]

硕士研究生 (协助指导)

2024 – 2027 聂海亮 (新疆大学) [在读]; 张美芳 (新疆大学) [在读]; 杨鑫云 (新疆大学) [在读];
李晶 (石家庄铁道大学) [在读]

2023 – 2026 邱燕萍 [毕业]; 赵亚茹 [2025 年硕博连续];
郭长明 (新疆大学) [获硕士研究生国家奖学金, 2026 年硕博连续]

2022 – 2025 宋宜璇 (新疆大学) [毕业]; 汤子钰 [毕业]

2021 – 2024 王彩凤 [德州审计局]; 李嘉乐 [德州统计局]; 郭爽 [泰安市中行];
刘馨茹 [连云港税务局]; 刘云霞 (新疆大学) [北京某科研机构]

2020 – 2023 付青城 [获硕士研究生国家奖学金、校优秀硕士学位论文, 去南方科技大学数学系读博，师从 Alexander Kurganov 教授]

2019 – 2022 李闰 [获硕士研究生国家奖学金, 重庆行知学校];
李婷婷 (石家庄铁道大学) [北京通州运河中学]

2017 – 2020 李东梅 [青岛体育运动学校]; 方丽丽 [滨州财政局]; 徐捷 [舟山中学]; 杨洋 [毕业]

本科生

2021 – 2025 俞磊, 留在学院读研，师从付凯副教授; 于子峻

关于研究生

- 导师实力：吾虽无教授之职称，但有教授之实力，办公室虚席以待学生讨论
- 学生待遇：适量劳务费，固定自习位置，高性能电脑，老师不抢学生一作
- 期待学生：热爱科研坐得住，刻意练习多行动，直说问题多交流
- 如果你赞成下面这句话，那么欢迎你，我的伙伴！

业精于勤，功不唐捐！